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In-situ microaeration of anaerobic digester treating buffalo manure for enhanced biogas yield

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ABSTRACT

Microaeration pretreatment can enhance lignin degradation of lignocellulosic biomass and subsequent biogas production during anaerobic digestion (AD). However, the reported values of effective aeration dose, aeration contact time, and aeration frequency differ with substrates riddling the selection of a suitable combination. A serial approach was adopted to identify the suitable combination for the degradation of lignin in the buffalo manure and improvement of the biogas production. Single and Multiple injections of three aeration doses of 5.3, 7.3, and 9.3 mL O_2/g VS were tested at 5 different contact times− 6, 12, 24, 36, and 47 h. Results for acid-soluble lignin (% ASL) indicated the suitability of 12 h contact time with a single injection of 7.3 mL O_2/g VS for lignin degradation and biogas enhancement for AD of buffalo manure. This combination achieved 78% higher ASL content and a 32% increase in average daily biogas yield than the control. Finally, the kinetic parameters obtained through the modified Gompertz model accurately predicted the biogas potential of the process ($R^2 > 0.93$). The results of this study can be referenced for improving the yield of similar biomass to bioenergy initiatives by employing a low-cost pretreatment strategy of *in-situ* microaeration.

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1. Introduction

Animal manure laden with pollutants including antibiotic residues, hormones, and heavy metals is produced in abundance in developing countries [1]. A large amount of this waste is left untreated resulting in environmental pollution as well as a loss of a valuable resource [2]. Anaerobic digestion (AD) can effectively reduce the environmental harmfulness of organic waste [3] and simultaneously produce the much-needed bioenergy in the form of biogas [4]. Animal waste, even after the loss of nutrients in the animal gut, has enough nutrients that make it a low-cost input with valuable outputs through the AD process [5]. However, being lignocellulosic, the presence of lignin renders low efficiency to the AD of lignocellulosic biomass [6]).

Lignocellulosic biomass is composed of 20–50% cellulose, 15–35% hemicellulose, and 18–35% lignin [7]. During fermentation, the major portion of sugars is derived from the cellulosic part of lignocellulosic biomass, which is then converted into fuels and chemicals [8]. Both cellulose and hemicellulose interwound to form a network of fibers where lignin gets impregnated to form strong

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cemented linkages in biomass. Lignin is a complex polymer made of aromatic alcohols (phenols) that makes it difficult to hydrolyze during AD [9]. Thus, the limited biodegradation of lignocellulosic biomass is one of the key challenges to AD [10].

The first step in the multi-step AD process is the hydrolysis of polymers [11]. For lignocellulosic biomass, hydrolysis is the rate-limiting step for the AD process [12]. Thus animal wastes containing higher lignin content will give limited bioenergy through conventional AD [13] due to the high content of hardly degradable lignocellulosic content in the feedstock [14]. Application of a suitable biomass pretreatment method for lignin removal before AD can, therefore, enhance the efficiency and increase the biogas production [15]. Table 1 summarizes a variety of biomass pretreatment methods– physical, chemical, and biological— that are commonly practiced these days.

It can be seen in Table 1 that each pretreatment method has its pros and cons and, no single method is advantaged over the other. For example, physical methods cannot disrupt the lignocellulosic structure, chemical processes require complex technical knowledge for smooth operation or else they will pose a risk to the process and the equipment, and finally, the biological processes are too slow to be adopted. Thus, choosing an effective pretreatment method is not an easy task. Recently, microaeration of the substrate for disrupting the lignocellulosic structure is reported as an effective pretreatment method requiring minimal financial and technical resources (Cesaro & Belgiorno, 2014; [8,19].

The introduction of small amounts of oxygen in the anaerobic digester has been described as a low-cost and effective strategy to promote the hydrolysis efficiency of the process [20,21]. Injecting small amounts of oxygen (microaeration) in the anaerobic digester is found to be an effective *in-situ* biomass pretreatment method that facilitates the growth of microbes and stimulates hydrolysis of complex organic matter [8]. Initially, microaeration was used for biogas up-gradation by desulphurization of biogas, but now it is also utilized for improving the hydrolysis and methanogenesis step of AD [22]). [23]; found positive effects of micro-aerated anaerobic digestion system on facultative microbes such as *Firmicutes* and *Clostridia* resulting in a better hydrolysis rate. Similarly, the methanogenic step was also improved because of the prevalence of aerotolerant *Methanosarcina* and *Methanobacterium* spp. in the microaerated anaerobic digester.

The positive effects of the microaeration pretreatment to achieve enhanced hydrolytic rates during the AD process have been reported AD of wastewater [24,25], black water [26], food waste [19,27], energy crops (Kaparaju et al., 2009), and agricultural residues [8,28,29]. Previous literature identifies three common variables– aeration dose, aeration contact time, and aeration frequency that play the dominant role in the effectiveness of microaeration pretreatment for an efficient AD process. However, the diversity of substrates and the associated disparity of the suitable parameters used in available literature makes it challenging to achieve the positive effects of microaeration [30]. Another important yet understudied effect of microaeration pretreatment is on lignin degradation. In the case of lignocellulosic biomass, hydrolysis is the rate-limiting step, and the efficiency of a pretreatment process is significantly dependent on the degradation of lignin. The degradation of lignin significantly improves the availability of sugars for microbial consumption which ultimately leads to better biogas yield. Finally, administering an improper amount of oxygen can be inhibitory rather than beneficial for the methanogens [21]. Thus, the availability of congruous data can increase the adaptability of the low-cost *in-situ* pretreatment strategies like microaeration particularly in low-income countries where cost is the governing factor in decision making.

AD of buffalo manure is one of the most feasible bioenergy production processes in developing countries like Pakistan. Buffalo is a common domestic animal in Pakistan reared for milk and meat and its waste is often added to fields as fertilizer. In few instances where AD is practiced in Pakistan, the primary substrate is buffalo waste. It is estimated that there is a potential of 16.3 million m^3 per day biogas production in Pakistan that can cater to the energy needs of almost 112 million residents of rural areas [31]. However, quantitative fluctuation in gas production has largely hindered the adoption of this process in Pakistan [32]. Therefore, finding a cost-effective and technically simple biomass pretreatment method is imperative to enhance the biogas yield from the manure AD process and to increase its popularity and adaptability in developing countries. This study is aimed at determining the specific aeration dose, contact time, and aeration frequency for efficient degradation of lignin present in the buffalo manure for subsequent improvement of biogas production during the AD process.

2. Materials and methods

2.1. Substrate and inoculum

Fresh 'as excreted' buffalo manure was collected and used as the substrate for this research. Fresh manure samples were taken from nearby farms in district Abbottabad. After collection, the substrate was stored at 4 °C until used in the experiment. The inoculum used in this study was taken from already operating lab scale digesters. Before its use in the experiments, the inoculum was stored under anaerobic conditions until the gas production stopped. The physicochemical characteristics of substrate and inoculum are shown in Table 2.

Table 2
Characterization of substrate and inoculum.

Parameters	Unit	Substrate	Inoculum
Total solids	%	14.6 ± 0.001	4.16 ± 0.008
Volatile solids	%	12.26 ± 0.047	3.96 ± 0.010
TKN	g/L	1.45 ± 0.04	n.d
P measured as P_2O_5	mg/L	249.95 ± 14.33	95.626 ± 6.20
COD	g/L	68.14 ± 7.19	10.836 ± 2.59
pH	–	7.86 ± 0.21	7.26 ± 0.42
Acid soluble lignin	%	17.09	n.d

Table 1
Different methods for pretreatment.

Method	Example	Purpose	Pros	Cons	Reference
Physical	Milling; Irradiation; Microwave; streaming	Increase in accessible surface area	Hemicellulose hydrolysis	High energy demand; Lignin not removed	[16]) [17]
Chemical	Acid/Alkali treatment; Organic solvent	Delignification and hydrolysis of hemicellulose	An effective and rapid method	Corrosion of the digester; Costly method	[16]) [17]
Biological	Microbes/enzymes	Hydrolysis, delignification	Increased bioavailability of substrate; Effective method for enhancing bioenergy	Low treatment rate	[16–18]

2.2. Experimental setup

The whole experiment was divided into two phases to achieve the overall objective of the study.

2.2.1. Phase-I

Phase-I of this study was used to screen out the best aeration dose, aeration contact time, and frequency of aeration. Initial values for these parameters were selected based on the available literature. The outcome of this phase was measured as the decrease in the amount of acid-soluble lignin (ASL) as compared to control. Accordingly, three sets of screening experiments were performed. In each experiment, 10 g of the substrate was added in a 60 mL glass bottle and flushed with nitrogen for up to 3 min, and sealed immediately to make it airtight. Afterward, the gases were evacuated, and a specified amount of air (as mentioned in Table 3) was injected into the digesters. The treatments were run in triplicates and the performance was measured against control bottles containing unaerated substrates.

During experiment 1, the efficacy of 7.3 mL O_2/g VS aeration dose was tested at 5 levels of time— 6 h, 12 h, 24 h, 36 h, and 47 h to select the best contact time for lignin degradation. The initial selection of 7.3 mL O_2/g VS aeration dose was based on the previous tests conducted with this dose using a lignocellulosic substrate with a similar biochemical composition [23,29]. After identifying a suitable contact time for aeration, experiment 2 tested the efficacy of three levels of aeration doses— 5.3 mL O_2/g VS, 7.3 mL O_2/g VS, and 9.3 mL O_2/g VS at two levels of time— 12 h and 24 h to select the best aeration dose for lignin degradation. Finally, experiment 3a and experiment 3b further elucidated the effect of pulse aeration on lignin degradation at 12 h and 24 h aeration contact time. The summary of the experimental design and conditions for the experiments are shown in Table 3.

2.2.2. Phase-II

Phase-II of this study aimed at finding the effectiveness of aeration pretreatment on enhancing biogas yield during batch anaerobic digestion of the buffalo manure. Manure samples added in triplicate 1-L bottle-reactors (600 mL effective volume) were first aerated with a single aeration dose of 7.3 mL O_2/g VS for 12 h (based on the findings of screening experiments: phase-I). Afterward, the anaerobic inoculum was added to each digester to achieve a substrate to inoculum ratio of 1 on a volatile solid basis. All the reactors were flushed with nitrogen for 3–5 min to create anaerobic conditions and sealed immediately. The digesters were placed in a water bath at 35 ± 2 °C for 14 d to mimic the mesophilic AD process. The digestion was stopped at day 14 since the data for biogas production for both the control and treatments did not show any significant change in daily biogas production beyond day 10. The digesters were manually shaken for 10 min every day to obtain thorough mixing. Following the same protocol, triplicate 1-L bottle-reactors containing unaerated manure samples and inoculum were used as control. Background biogas produced by the inoculum was measured under the same conditions and adjusted for the other biogas digesters.

2.3. Analytical methods

Total solids, volatile solids, chemical oxygen demand, pH, and NH_4–N of the substrate and inoculum were measured according to the standard methods (APHA, 1998). The standard NREL method (National Renewable Energy Laboratory) was used for the determination of lignin content in lignocellulosic biomass [35]). The spectrophotometric method was used to estimate total phenolic content (TPC) as explained by [36]. Briefly, 1 mL extract was mixed with 5 mL of diluted Folin Ciocalteu followed by mixing of 4 mL Na_2CO_3 solution (7.5%). The mixture was incubated at room temperature in dark for 90min and absorbance was measured at 760 nm. Measured levels of TPC were stated as mg Gallic acid equivalents in 100g of the fresh sample (mg GAE/100g, FW). Biogas produced in each digester was removed daily with the help of a 50 mL syringe and quantified. The measured wet biogas volume was then adjusted for standard temperature (0 °C) and pressure (1atm) according to equation (1) as described in [37].

$$V_{stp} = V_T \times 273 \frac{760 - \rho_w}{(273 + T) \times 760} \quad (1)$$

where V_{STP} is the volume of gas measured at standard temperature (0°) and pressure (1atm), V_T is the volume of gas measured at given temperature T, and ρ_w is the vapor pressure of the water as a function of temperature (mm Hg).

2.4. Statistical analysis

The outputs of the experiments were statistically analyzed by single-factor analysis of variance (1-way ANOVA) using Minitab software version 16 at $\alpha = 0.05$. When the results of ANOVA showed a significant difference, the means were further compared using *Post-hoc* Tukey's test [37].

2.5. Kinetic study

The reliability of the experimental data obtained for cumulative biogas yield is widely ensured by fitting the data into the Gompertz kinetic model [38]. Thus, a modified Gompertz model (Equation (2)) was used in this study to fit the kinetics of biogas production during batch AD of aerated and un-aerated buffalo manure. The simulation results provide insightful information about the AD process by predicting lag time, biogas production potential, technical digestion time, and maximum biogas production rate of the anaerobic digestion process.

$$M = P \times \exp\left\{ -\exp\left[R_m \exp\frac{\lambda - t}{P} + 1 \right] \right\} \quad (2)$$

where M is the cumulative biogas production (mL) at a given time t (d); P is the biogas production potential (mL); R_m is the maximum biogas production rate (mL/d); and λ is the lag phase time (d), which represents the time taken by the bacteria to acclimate to the digester conditions before gas production starts. The best values of

Table 3
Summary of experimental details for phase I.

Experiment	Aeration dose (mL O_2/g VS)	Aeration contact time (h)	Aeration frequency	Purpose of experiment	References followed
1	7.3	6,12, 24, 36, 47	Single	Identification of the best contact time	[23,29]
2	5.3, 7.3, 9.3	12, 24	Single	Identification of best aeration dose	[29]
3a	7.3	12	Multiple (aeration at 0h, 1h, 6h)	Identification of best aeration frequency	[33,34]
3b	7.3	24	Multiple (aeration at 0h, 1h, and 12h)	Identification of best aeration frequency	[33,34]
Control	–	0	–	Compare treatment performance	Current study

P, R_m, and λ, in our study, were determined using Microsoft Excel™ "SOLVER" tool by minimizing the sum of the square of errors (SSE) between experimental and model-predicted data [39].

3. Results and discussion

3.1. Influence of microaeration pretreatment on lignin degradation

For phase I, a sequence of three screening experiments was conducted to identify the best combination of aeration contact time, aeration dose, and aeration frequency to achieve maximum lignin degradation measured as percent acid-soluble lignin (% ASL) of the substrate biomass. During experiment 1, the substrate was aerated at five levels of contact time while injecting all digesters with 7.3 mL O_2/g VS of the substrate to identify the aeration contact time that achieves maximum lignin degradation of the substrate. Fig. 1 shows the results of experiment 1 of phase I for lignin degradation in the substrate.

It can be seen in Fig. 1 that the digester injected with 7.3 mL O_2/g VS of the substrate and 12 h contact time contained only 3.96% acid-soluble lignin (ASL) degrading 78% more ASL as compared to control (initial ASL content of 17.09%) followed by 77% ASL degradation at 24 h, 68% ASL degradation at 36 h, 60% ASL degradation at 47 h, and only 4.5% ASL degradation at 6 h. Thus, based on this finding 12 h was identified as the best aeration contact time for delignification using a single injection of 7.3 mL O_2/g VS of the substrate. Furthermore, to identify the best aeration dose, experiment 2 of phase I was carried out with 24 h and 12 h of aeration contact time. Since the results of lignin degradation achieved at 12 h and 24 h were not significantly different ($\alpha = 0.05$) both contact times were further tested in experiment 2. Previous studies have also reported the suitability of 12 h and 24 h aeration contact time for biomass aeration pretreatment and subsequent improvement in biogasification through anaerobic digestion [23,33,34].

Fig. 2 shows the results of experiment 2 of phase I where three aeration doses—5.3 mL O_2/g VS, 7.3 mL O_2/g VS, and 9.3 mL O_2/g VS of the substrate were tested for their delignification ability.

The results revealed that aeration dose of 7.3 mL O_2/g VS dose was most effective in lignin degradation among three tested aeration doses. Accordingly, 7.3 mL O_2/g VS aeration dose resulted in higher lignin degradation for 12 h aeration contact time compared to 24 h. For 12 h aeration contact time, 7.3 mL O_2/g VS dose resulted in 3.96% ASL content which was 68% lower than the 5.3 mL O_2/g VS dose (12.32% ASL) and 75% lower than the 9.3 mL O_2/g VS dose (15.61% ASL). Similarly, for 24 h aeration contact time, 7.3 mL O_2/g VS dose resulted in 4.12% ASL content which was 71% lower than the 5.3 mL O_2/g VS dose (14.31% ASL) and 65% lower than the 9.3 mL O_2/g VS dose (11.93% ASL). Thus, the results of experiment 2 depict

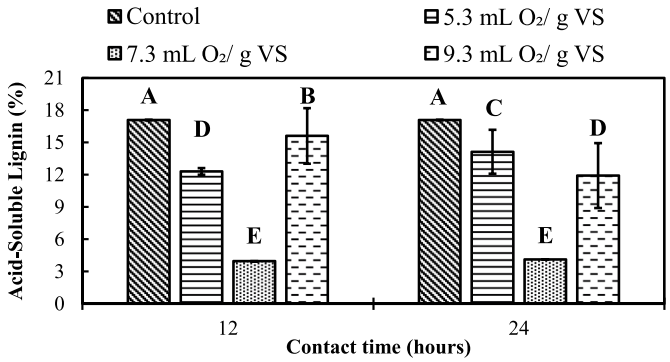


Fig. 2. Effect of aeration dose on lignin degradation at 12 h and 24 h contact time. Different letters on top of the bars show significant difference among the treatments ($p < 0.05$ at $\alpha = 0.05$).

that 12 h of aeration contact time with a single injection of 7.3 mL O_2/g VS aeration dose caused maximum degradation of lignin. Finally, experiment 3 was conducted to study the effect of aeration frequency on lignin degradation. For this, multiple doses of 7.3 mL O_2/g VS were injected in the respective digesters during 12 h and 24 h contact times (Table 3). For single aeration, 7.3 mL O_2/g VS was injected in the respective digesters at the beginning of the experiment. Whereas, for multiple aerations, 7.3 mL O_2/g VS was injected in the respective digesters at 0 h, 1 h, and 6 h during 12 h contact time, and for 24 h contact time 7.3 mL O_2/g VS was injected at 0 h, 1 h, and 12 h.

Fig. 3 shows the results of experiment 3 of this study.

It is clear in Fig. 3 that increasing the frequency of aeration did not significantly affect the lignin degradation both at 12 h and 24 h contact times. For 12 h contact time, three doses of 7.3 mL O_2/g VS air were injected in the respective digesters at 0 h, 1 h, and 6 h that resulted in 16.91% ASL which was not significantly different than control (17.09% ASL). Similarly, for 24 h contact time, three doses of 7.3 mL O_2/g VS air were injected in the respective digesters at 0 h, 1 h, and 12 h that resulted in 16.89% ASL which was again not significantly different than control (17.09% ASL). Thus, injection of a single dose of 7.3 mL O_2/g VS for 12 h and 24 h was deemed suitable to degrade the lignin in the substrate resulting in 3.96% ASL for 12 h contact time and 4.12% ASL for 24 h contact time.

[33] suggested improved hydrolysis of lignocellulosic biomass when microaerobically pretreated before the anaerobic digestion (AD). Hydrolysis of biomass occurs naturally both under aerobic and anaerobic conditions, but in the case of aerobic start, the production of extracellular hydrolytic enzymes by facultative hydrolytic microbes increases by manifolds [40]. The presence of oxygen

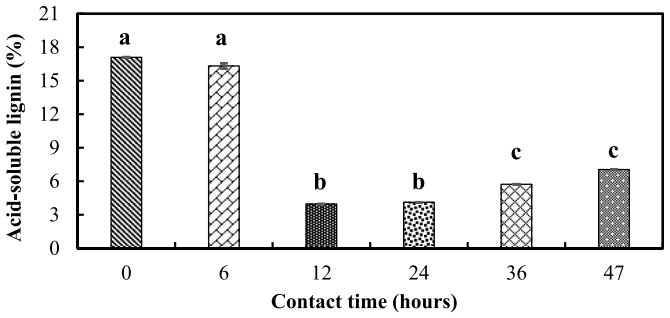


Fig. 1. Effect of aeration contact time on lignin degradation. Different letters on top of the bars show statistically significant difference among the treatments ($p < 0.05$ at $\alpha = 0.05$).

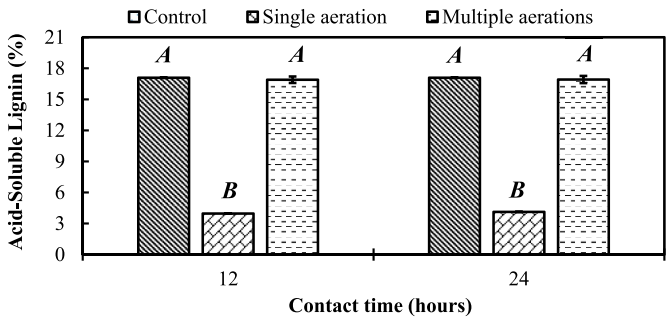


Fig. 3. Effect of aeration frequency on lignin degradation at 12 h and 24 h contact time. Different letters on top of the bars show significant difference among the treatments ($p < 0.05$ at $\alpha = 0.05$).

promotes lignin oxidation due to microbial enzymes, which can break down the polyphenolic polymer [20] and significantly improve the downstream biomethanation [29]. Thus the establishment of microaerobic conditions in the anaerobic digestion process seems to be an appropriate solution that can improve the hydrolysis step and boost biogas yield from lignocellulosic biomass [41].

[29] found that the injection of two pulses 7.3 mL O_2/g VS of aeration on agricultural residues enhanced biomass hydrolysis and resulted in a 9% enhanced biomethane yield for a treatment period of 47 h. While increasing the dose to 15 mL O_2/g VS hurt hydrolysis and thus decreased methane yield. The optimization results of the present study indicate that digester injected with a single injection of 7.3 mL O_2/g VS of the substrate for 12 h contact time degraded 78% acid-soluble lignin (ASL) and resulted in 32% enhancement of the biogas yield.

The literature suggests the employment of microaeration for short periods to achieve efficient hydrolysis of organic material at the beginning of the digestion process [33]. This would not only enhance the hydrolysis of organic biomass but also promote the acidogenesis step. On the contrary, a higher dosage of oxygen for an extended period can inhibit methanogenic activity and gas yield [21]. Accordingly, in the present study 7.3 mL O_2/g VS aeration dose resulted in higher lignin degradation for 12 h followed by 24 h aeration contact times. For 12 h aeration contact time, 7.3 mL O_2/g VS dose resulted in 68% higher lignin degradation than the 5.3 mL O_2/g VS dose and 75% higher than the 9.3 mL O_2/g VS dose. Similarly, for 24 h aeration contact time, 7.3 mL O_2/g VS dose resulted in 71% higher lignin degradation than the 5.3 mL O_2/g VS dose and 65% higher than the 9.3 mL O_2/g VS dose. Thus, in addition to the suitable aeration dose (described in previous studies), the current study also suggested suitable aeration contact time and aeration frequency for the degradation of lignin.

3.2. Influence of microaeration pretreatment on biogas production

Based on the lignin degradation data collected in phase I, the buffalo manure was pretreated with 7.3 mL O_2/g VS aeration dose for 12 h before phase II (anaerobic digestion) of this study. Pretreated buffalo manure was anaerobically digested at mesophilic conditions (35 °C) for 14 d. Fig. 4a compares the daily biogas yield of pretreated (aerated) and untreated (unaerated) manure samples. In both cases, the biogas production started on day 1. However, the biogas yield of the aerated manure samples remained higher than the unaerated manure samples for the first 10 days of the digestion. On average the daily biogas yield for the pretreated samples was 79 ± 9 mL biogas/g VS/d which was approximately 32% higher than the control (54 ± 2 mL biogas/g VS/d). The biogas production flattened for both the treated samples and control after 10 d and did not change significantly thereafter. Overall, during 14 d digestion, there was an accumulation of 1226 ± 53 mL of biogas in the treated digesters whereas the untreated digesters accumulated only 900 ± 12 mL biogas (27% lesser than treated samples) as shown in Fig. 4b.

The effectiveness of microaerobic pretreatment for the promotion of hydrolysis and subsequent biogasification has been established by the literature. [29]; reported that biomethane production increases by 9% with the injection of 7.3 mL O_2/g VS aeration dose for microaerobic pretreatment. Another study reported a 7% enhancement in anaerobic digestion with the injection of 7.3 mL O_2/g VS [23]. Similarly, it was reported that the highest enhancement in biogas production in wheat straw was obtained with microaeration pretreatment of 5 mL O_2/g VS [29]. However, the effective aeration dose varies with the substrate. Table 4 shows the effects of aeration doses and aeration contact times on the hydrolysis and biogas yield of different substrates reported in the literature.

It can be seen in Table 4 that aeration can have both positive and negative effects on the biogas yield. For instance, the agricultural residues pretreated with 7.3 ml O_2/g VS/day had a 7% enhancement in biogas production while increasing the aeration dose to 15 ml O_2/g VS/day negatively impacted the process [23]. Similarly, insufficient microaerobic conditions showed decreased performance in biogas yield. This might be due to hampering in production of the enzymes for hydrolysis of biomass [33]. [28] reported that the maximum methane yield was obtained with microinjection of O_2

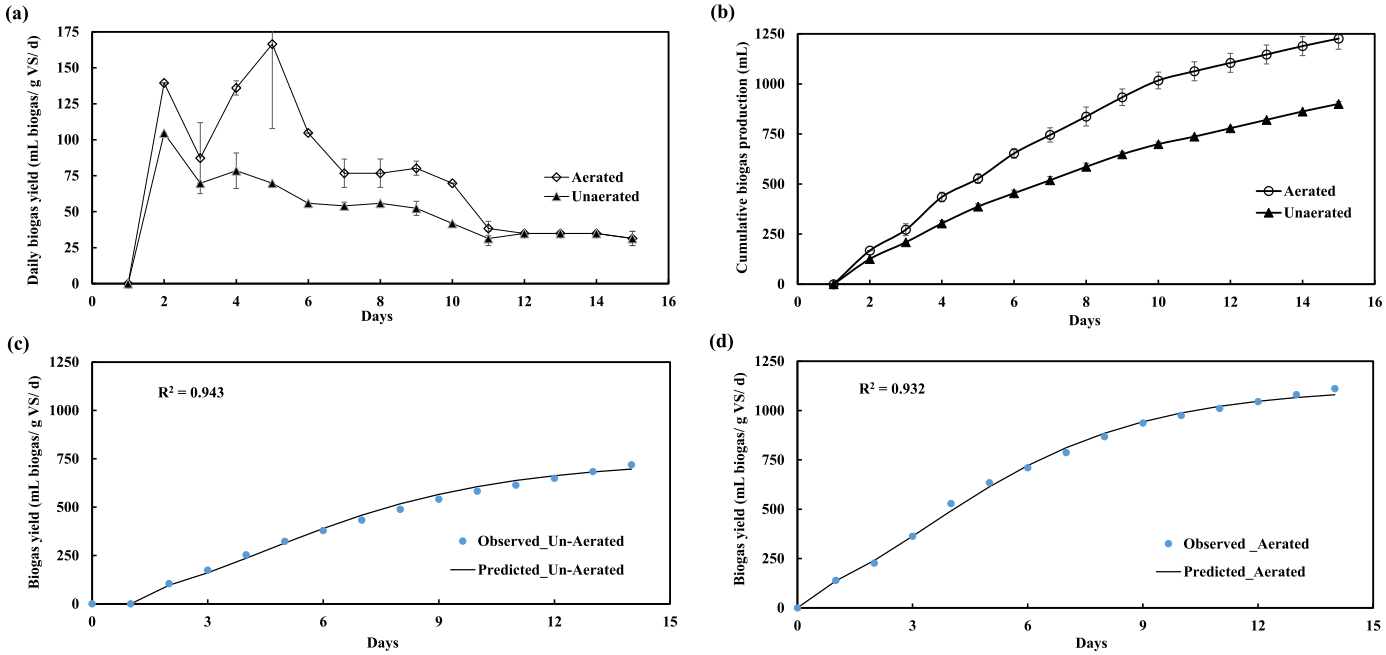


Fig. 4. Biogas production and kinetics of aerated and unaerated buffalo manure samples, (a) daily biogas yield, (b) cumulative biogas production, (c) Gompertz predicted yield of unaerated substrate, (d) Gompertz predicted yield of aerated substrate.

Table 4
Summary of literature carried on microaeration pretreatment that affects the performance of AD.

Biomass	Aeration dose	Contact time	Effect on hydrolysis	Effect on gas yield	References
Brown water	37.5 mL $O_2/L_R/d$	4 d	Positive impact	21–23%	[19]
Food waste	37.5 mL $O_2/L_R/d$	4 d	Positive impact	10%	[19,40]
	5 L/h	24 h		6%	
Corn straw	12.5 mL $O_2/L_R/d$	n.d	10.3%	16.5%	[20]
Corn straw	5 mL O_2/g VS	n.d	Positive impact	16.2%	[20]
Corn straw	7.3 mL O_2/g VS	47 h (2 pulses)	Positive impact	9%	[29]
Agricultural residues	5 mL O_2/g VS	n.d	Positive impact	7.2%	[23]
	7.3 mL O_2/g VS		Positive impact	7%	
	15 mL O_2/g VS		Negative impact	−4%	
Sewage sludge	0.3 vvm	48 h	36.2% decrease in volatile suspended solids	211%	[25]
Buffalo manure	7.3 mL O_2/g VS	12 h	Positive impact	32%	Current study

n.d* represents no data.

Table 5
The kinetic parameters of the study measured through the Gompertz model.

Treatment	$T_{90\ (days)}$	lag time (days)	Predicted biogas potential (mL/g VS/d)	Observed biogas potential (mL/g VS/d)	Observed vs Predicted difference (%)
Un-Aerated Substrate	12	0	1080 ($R^2 = 0.943$)	1112	3
Aerated Substrate	11	0	708 ($R^2 = 0.932$)	750	6

with a dosage of 5 ml O_2/g VS, which was found to be 17.35% higher than control. In the current study, we witnessed similar positive results of injecting 7.3 ml O_2/g VS at the beginning of digestion with the accumulation of 27% more biogas in total and 32% more biogas yield on daily basis as compared to the untreated substrates. Thus, the findings of the present study agreed with the previously reported improvements in biogasification due to aerobic pretreatment.

The kinetic analysis performed for this study also supported the positive role of microaeration on the biogas potential of the process (Fig. 4c and d). For example, the time taken by aerated samples to give off 90% biogas potential was 24 h less than that of the unaerated samples (Table 5). This reduction in digestion time can have positive economic and environmental impacts such as reduced digester footprint.

3.3. Total phenolics production, pH, total and volatile solids reduction

Lignin is a complex polyphenolic biopolymer, and its degradation leads to the production of phenolic compounds [42]. Accordingly, the concentration of total phenolics was measured to verify the degradation of lignin due to aeration pretreatment as compared to control. As illustrated in Fig. 5a, the concentration of total phenolics of the microaerobically pretreated substrate was higher than the control. This increase in total phenolics during aerobic pretreatment was found to have a positive effect on biogas yield during anaerobic digestion as described in previous studies [43]. Another important parameter that controls the performance of the AD process is pH with an optimum range of 6–8.5 [44]. For the current study, the pH of both the aerated and unaerated substrate remained within the suitable range as shown in Fig. 5b. Buffalo manure

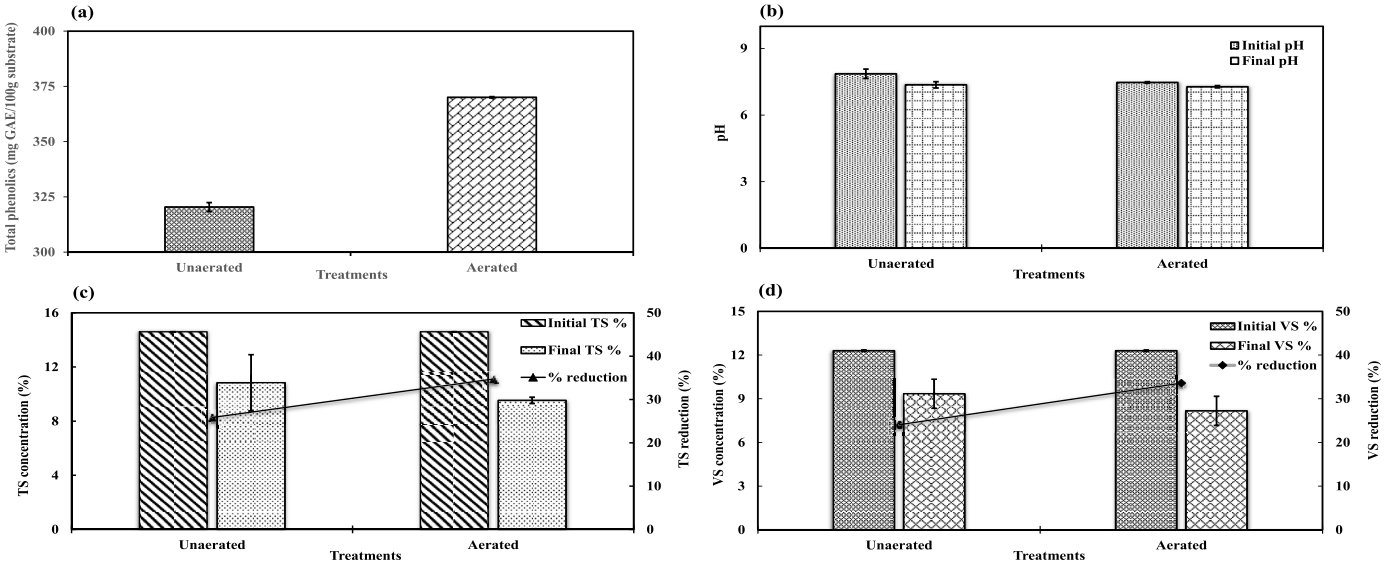


Fig. 5. (a) Total phenolic content (b) pH, (c) TS reduction, and (d) VS reduction of control and the aerated substrate.

carries a strong buffering capacity which makes it a suitable substrate for the AD process. Further, the aeration pretreatment results in the removal of ammonia which is a possible reason for the lower pH of the aerated substrate as compared to the control [45]. Finally, the performance of the AD process can be measured by its ability to volatilize the available solids into biogas [2]. It can be seen in Fig. 5c and d that both the total and volatile solids reduction was higher in aerated treatment as compared to the control. This higher reduction of the solids during AD of the treated substrate was reflected in higher biogas yield.

4. Conclusions

The current study demonstrated that a suitable combination of aeration dose, aeration frequency, and aeration contact time effectively enhanced the lignin degradation and thus the biogas yield ($p < 0.05$). A single injection of 7.3 ml O_2/g VS of the substrate for 12 h contact time degraded 78% more acid-soluble lignin and achieved a 32% increase in biogas yield on daily basis as compared to the control. Increasing the aeration frequency or aeration dose beyond 7.3 ml O_2/g VS did not improve the lignin degradation. The modified Gompertz model effectively predicted the kinetic parameters associated with biogas production under the given experimental conditions process ($R^2 > 0.93$). The findings of this study advocate the use of a low-cost microaerobic pretreatment strategy for the promotion of potential biomass to bioenergy projects in developing countries like Pakistan.

Credit authorship statement

S. Yousaf: Writing - original draft, Visualization, Biochemical analyses, **A. Yasmeen**, **A. Zeb**, **J.A. Tariq, M. Bilal**, and **A.M. Abbasi**: Biochemical analyses and data visualization **M. Ali, T.M. Khalil, A. Yaqoob, and R. Ahmed:** Writing - review & editing, Visualization, Resource, **Q. Zhao:** Investigation, Visualization, Writing - review & editing **I. Zeb:** Conceptualization, Investigation, Visualization, Writing - review & editing, Resources, Supervision.

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Credit authorship contribution statement

Iftikhar Zeb: Conceptualization, Investigation, Visualization, Writing – review & editing, Resources, Supervision. **Sana Yousaf:** Writing – original draft, Visualization, Formal analysis. **Muhammad Ali:** Writing – review & editing, Visualization, Resources. **Aqsa Yasmeen:** Formal analysis, and data, Visualization. **Anwar Zeb Khan:** Formal analysis, and data, Visualization. **Junaid Ahmad Tariq:** Formal analysis, and data, Visualization. **Quanbao Zhao:** Investigation, Visualization, Writing – review & editing. **Arshad Mehmood Abbasi:** Formal analysis, and data, Visualization. **Raza Ahmad:** Writing – review & editing, Visualization, Resources. **Tariq Mahmood Khalil:** Writing – review & editing, Visualization, Resources. **Asim Yaqoob:** Writing – review & editing, Visualization, Resources. **Muhammad Bilal:** Formal analysis, and data, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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